

Log-Concavity of Domination Polynomials for a Seven-Parameter Class of Three-Clique Split Graphs

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Abstract

Let $C = \{1, 2, 3\}$ induce a clique. For every nonempty $T \subseteq C$, add a_T independent vertices whose neighborhood is exactly T , where the seven multiplicities a_T are arbitrary nonnegative integers. We prove that the coefficient sequence of the domination polynomial of every graph in this class is log-concave. The proof first reverses the coefficients and embeds the resulting polynomial in a larger six-exponent family of shifted binomial polynomials. A strengthened coefficient inequality for that family is then proved by induction. The finite ingredients are exhaustive exact-integer checks of the induction base, a bounded endpoint reduction, and the original graph profiles of total multiplicity at most six.

1 Introduction

For a finite graph G , a set $S \subseteq V(G)$ is dominating if every vertex outside S has a neighbor in S . Its domination polynomial is

$$D(G, x) = \sum_{k=0}^{|V(G)|} d_k(G) x^k,$$

where $d_k(G)$ is the number of dominating k -subsets. A finite nonnegative sequence (c_k) is log-concave when $c_k^2 \geq c_{k-1}c_{k+1}$ at every internal index, with coefficients outside its displayed range understood to be zero.

Omar [1] proves unimodality for domination polynomials of threshold graphs using a total-positivity argument for shifted binomial rows, and identifies extension to split graphs after the nested-neighborhood condition is relaxed as a natural direction. The graph class below is a fixed three-vertex clique-side slice of that direction. It allows all seven nonempty neighborhood types with unrestricted repetition, so the independent-side neighborhoods need not be nested. The conclusion here is the stronger coefficient property of log-concavity, but only for this specified slice; it is not a result about all split graphs.

Exact coefficient computation also appears in other enumerative probability inequalities; see, for adjacent computational methodology only, Zeng [2]. That paper is not a source for any domination-polynomial assertion used here.

Let $C = \{1, 2, 3\}$. For each nonempty $T \subseteq C$, let I_T be a set of $a_T \geq 0$ vertices. Make C a clique, make $I = \bigsqcup_{\emptyset \neq T \subseteq C} I_T$ independent, and join a vertex of I_T precisely to the vertices in T . Denote the resulting graph by $G(\mathbf{a})$, and put $m = |I| = \sum_T a_T$. Empty neighborhood types are not part of the class. The specified clique need not be maximum; for example, vertices of type C can form a K_4 with it.

Our main result is the following all-parameter statement.

Theorem 1.1. *For every seven-tuple $(a_T)_{\emptyset \neq T \subseteq \{1,2,3\}}$ of nonnegative integers, the coefficients of $D(G(\mathbf{a}), x)$ satisfy*

$$d_k(G(\mathbf{a}))^2 \geq d_{k-1}(G(\mathbf{a}))d_{k+1}(G(\mathbf{a})) \quad (1 \leq k \leq m+2).$$

The infinite part of the proof is algebraic. Exact finite computations, whose domains and coefficient formulas are recorded in Section 4, provide its three bounded inputs.

2 Counting and coefficient reversal

For $A \subseteq C$, set

$$f(A) = \sum_{\substack{\emptyset \neq T \subseteq C \\ T \cap A = \emptyset}} a_T.$$

Let

$$\delta = \mathbf{1} \left\{ \bigcup_{a_T > 0} T = C \right\}, \quad s_i = a_{\{i\}}, \quad u_i = \sum_{T \ni i} a_T.$$

Lemma 2.1 (Counting identity). *The domination polynomial is*

$$D(G(\mathbf{a}), x) = \sum_{\emptyset \neq A \subseteq C} x^{|A|+f(A)} (1+x)^{m-f(A)} + \delta x^m. \quad (1)$$

Proof. Fix a dominating set S and first suppose that $A = S \cap C \neq \emptyset$. Every independent vertex of type T with $T \cap A = \emptyset$ must itself belong to S ; all other independent vertices are optional. Since C is a clique, the nonempty set A already dominates every clique vertex. This gives the summand indexed by A . If $S \cap C = \emptyset$, every independent vertex must belong to S , and this unique candidate dominates C exactly when the union of the occupied neighborhood types is C . This gives the last term. $\square$

Define $B_n(t) = (1+t)^n$. Reversing the coefficients in (1) yields the key reduction.

Lemma 2.2 (Reversal identity). *One has*

$$t^{m+3} D(G(\mathbf{a}), 1/t) = B_m(t) + t \sum_{i=1}^3 B_{m-s_i}(t) + t^2 \sum_{i=1}^3 B_{u_i}(t) + \delta t^3. \quad (2)$$

Proof. After reversal, the term indexed by $A \neq \emptyset$ in (1) becomes

$$t^{3-|A|} B_{m-f(A)}(t).$$

For $|A| = 3$, the binomial exponent is m . If $|A| = 2$ and i is the missing clique vertex, then $f(A) = s_i$. If $A = \{i\}$, then $m - f(A) = u_i$. Finally, the exceptional monomial reverses to δt^3 . $\square$

Coefficient reversal preserves every adjacent log-concavity inequality. It is therefore enough, for $m \geq 7$, to prove a slightly stronger result in which the six exponents in (2) vary independently.

3 An auxiliary shifted-binomial theorem

For integers $m \geq 7$, $0 \leq p_i, q_i \leq m$, and $\delta \in \{0, 1\}$, put

$$F_{m,p,q,\delta}(t) = B_m(t) + t \sum_{i=1}^3 B_{p_i}(t) + t^2 \sum_{i=1}^3 B_{q_i}(t) + \delta t^3, \quad r_k = [t^k] F_{m,p,q,\delta}. \quad (3)$$

Theorem 3.1. *For every choice of parameters in (3), the sequence (r_k) is log-concave.*

The proof uses the following stronger inequality for the $\delta = 0$ family:

$$r_k^2 - r_{k-1}r_{k+1} \geq r_{k+1} \quad (k \geq 3). \quad (4)$$

We first isolate its only index that is not propagated directly by the induction.

Lemma 3.2 (The index-three bound). *For $m \geq 7$, $0 \leq p_i, q_i \leq m$, and $\delta = 0$,*

$$r_3^2 - r_2r_4 \geq r_4.$$

Proof. Write

$$P = \sum_i p_i, \quad X = \sum_i \binom{p_i}{2}, \quad Y = \sum_i \binom{p_i}{3}, \quad Q = \sum_i q_i, \quad Z = \sum_i \binom{q_i}{2}.$$

Then

$$r_2 = \binom{m}{2} + P + 3, \quad r_3 = \binom{m}{3} + X + Q, \quad r_4 = \binom{m}{4} + Y + Z.$$

Fix p and two of the q -coordinates, and call the remaining coordinate n . The required margin, as a function of n , has the form

$$(B + n)^2 - (r_2 + 1) \left(C + \binom{n}{2} \right). \quad (5)$$

Its quadratic coefficient is $1 - (r_2 + 1)/2 < 0$. Thus its minimum on $[0, m]$ is attained at an endpoint. Applying this coordinatewise reduces the problem exactly to $q_i \in \{0, m\}$. By symmetry only the number $j \in \{0, 1, 2, 3\}$ of coordinates equal to m matters.

For $7 \leq m \leq 16$, exhaustive exact evaluation over all weakly sorted p -triples and all four values of j verifies 18,540 endpoint cases. The least margin is 196; further reproducibility details appear in Section 4.

It remains to treat $m \geq 17$. Since $P \leq 3m$ and

$$Y \leq \frac{m-2}{3}X,$$

putting $A = \binom{m}{2} + 3m + 4$ bounds the endpoint margin below by

$$X^2 + L_j(m)X + K_j(m), \quad (6)$$

where

$$L_j(m) = 2 \left(\binom{m}{3} + jm \right) - \frac{A(m-2)}{3},$$

$$K_j(m) = \left(\binom{m}{3} + jm \right)^2 - A \left(\binom{m}{4} + j \binom{m}{2} \right).$$

Table 1: Positive ascending coefficient vectors for the large- m bound.

j	$6L_j$	$144K_j$
0	(2430, 567, 42, 1)	(1126080, 1370412, 365716, 43299, 2635, 81, 1)
1	(2634, 579, 42, 1)	(756432, 1361316, 371968, 43827, 2647, 81, 1)
2	(2838, 591, 42, 1)	(470016, 1362012, 378508, 44355, 2659, 81, 1)
3	(3042, 603, 42, 1)	(266832, 1372500, 385336, 44883, 2671, 81, 1)

For completeness, Table 1 gives the coefficient vectors of $6L_j(17+h)$ and $144K_j(17+h)$, in ascending powers of $h = m - 17$.

Every displayed coefficient is positive. Because $h \geq 0$ and $X \geq 0$, (6) is positive, completing the proof. $\square$

Lemma 3.3 (Strong induction). *For every $m \geq 7$, every $p, q \in \{0, \dots, m\}^3$, and $\delta = 0$, inequality (4) holds at every $k \geq 3$.*

Proof. At $m = 7$, separate permutation symmetry in p and q leaves $\binom{10}{3}^2 = 14,400$ weakly sorted profiles. Exhaustive exact-integer evaluation verifies (4) for all seven potentially relevant indices $3 \leq k \leq 9$, a total of 100,800 inequalities. The least margin when $r_{k+1} > 0$ is 21; a zero margin occurs only in the zero tail.

Assume the assertion at $m - 1 \geq 7$. Define

$$p_i^- = \max(p_i - 1, 0), \quad q_i^- = \max(q_i - 1, 0),$$

and let $s_k = [t^k]F_{m-1,p^-,q^-,0}$. If α and β count the zero coordinates of p and q , respectively, Pascal's identity gives

$$F_{m,p,q,0}(t) = (1+t)F_{m-1,p^-,q^-,0}(t) - \alpha t^2 - \beta t^3. \quad (7)$$

Put $b_k = s_k + s_{k-1}$. For a nonnegative log-concave sequence with no internal zeros, adjacent-ratio monotonicity gives $s_{k-1}s_k \geq s_{k-2}s_{k+1}$. Direct expansion then yields

$$\begin{aligned} b_k^2 - b_{k-1}b_{k+1} &= (s_k^2 - s_{k-1}s_{k+1}) + (s_{k-1}^2 - s_{k-2}s_k) \\ &\quad + (s_{k-1}s_k - s_{k-2}s_{k+1}) \geq s_{k+1} + s_k = b_{k+1}. \end{aligned} \quad (8)$$

The sequence (s_k) has no internal zeros: B_{m-1} makes the coefficients through degree $m - 1$ positive, and any extension contributed by a shifted binomial row joins that interval.

For $k \geq 5$, the correction terms in (7) do not meet the adjacent triple at k , so (8) proves the desired bound. At $k = 4$, one has $r_3 = b_3 - \beta$, $r_4 = b_4$, and $r_5 = b_5$, hence

$$r_4^2 - r_3r_5 = (b_4^2 - b_3b_5) + \beta b_5 \geq b_5 = r_5.$$

The case $k = 3$ is Lemma 3.2. This closes the induction. $\square$

Proof of Theorem 3.1. It remains to check the first two indices and the exceptional term. Put $P = \sum_i p_i$, $X = \sum_i \binom{p_i}{2}$, and $Q = \sum_i q_i$. Then

$$r_0 = 1, \quad r_1 = m + 3, \quad r_2 = \binom{m}{2} + P + 3, \quad r_3 = \binom{m}{3} + X + Q + \delta.$$

As $P \leq 3m$,

$$r_2 \leq \binom{m+3}{2} < (m+3)^2 = r_1^2.$$

For the index-two defect, let $A = \binom{m}{2} + 3$. The bounds $X \leq (m-1)P/2$ and $Q + \delta \leq 3m + 1$ give

$$r_2^2 - r_1 r_3 \geq A^2 - (m+3) \left(\binom{m}{3} + 3m + 1 \right) + \frac{m^2 - 4m + 15}{2} P + P^2. \quad (9)$$

The coefficient of P is positive. With $h = m - 7$, the constant term in (9) is

$$\frac{h^4}{12} + \frac{11h^3}{6} + \frac{185h^2}{12} + \frac{140h}{3} + 6 > 0.$$

Lemmas 3.2 and 3.3 prove all remaining inequalities when $\delta = 0$. Passing from $\delta = 0$ to $\delta = 1$ increments only r_3 . This improves the defect at index three by $2r_3 + 1$, while it decreases the defect at index four by exactly r_5 , which is absorbed by (4). The index-two estimate already allowed $\delta = 1$, and all indices at least five are unchanged. Thus every adjacent defect is nonnegative. $\square$

4 Exact finite certificates

All finite calculations use exact integers (and exact rationals only for expanding the polynomials in Table 1); no floating-point comparison is involved. The coefficient formula used throughout is

$$r_k = \binom{m}{k} + \sum_{i=1}^3 \binom{p_i}{k-1} + \sum_{i=1}^3 \binom{q_i}{k-2} + \delta \mathbf{1}\{k=3\}, \quad (10)$$

with $\binom{a}{b} = 0$ outside $0 \leq b \leq a$.

The three bounded inputs are reproducible as follows.

1. For the original graph class with $m \leq 6$, enumerate all weak compositions of each m into seven labeled parts. There are $\sum_{m=0}^6 \binom{m+6}{6} = \binom{13}{7} = 1716$ profiles. For each profile, direct enumeration of all 2^{m+3} vertex subsets gives the domination coefficients. These agree with (1) and (2), and all 12,441 required adjacent inequalities hold.
2. For the $m = 7$ base of Lemma 3.3, enumerate the 120 weakly sorted p -triples and 120 weakly sorted q -triples, then evaluate (10) for $3 \leq k \leq 9$. This is exactly 14,400 profiles and 100,800 inequalities.
3. For Lemma 3.2 with $7 \leq m \leq 16$, enumerate every weakly sorted p -triple and $j = 0, 1, 2, 3$ after the proved endpoint reduction. This gives 18,540 cases. For $m \geq 17$, expand the explicit rational polynomials $L_j(17+h)$ and $K_j(17+h)$ to obtain Table 1.

The supplementary programs `triage-check.py`, `pass3-induction-check.py`, and the separately written `referee-check.py` implement these enumerations. The last program does not import or execute the first two and also performs direct graph subset enumeration on every $m \leq 6$ profile. Successful execution of a program is not substituted for any infinite argument above; the programs only discharge the explicitly listed finite domains.

5 Conclusion and scope

Proof of Theorem 1.1. For $m \geq 7$, set $p_i = m - s_i$ and $q_i = u_i$ in Theorem 3.1. These six exponents all lie in $[0, m]$, so the reversal identity (2) and coefficient reversal prove every claimed inequality. For $m \leq 6$, the complete direct enumeration in Section 4 supplies the remaining cases, including $m = 0$, when the graph is K_3 . $\square$

The theorem is confined to a specified three-vertex clique side and to nonempty independent-side neighborhood types. It neither proves unimodality for all split graphs nor asserts real-rootedness, strict log-concavity, or a classification of equality. The literature comparison made here is limited to the cited sources and is not a claim of priority.

Computational and AI assistance. OpenAI Codex assisted with proof exploration, organization of the exact finite checks, and preparation of the manuscript. The finite programs use the formulas and exhaustive domains stated in Section 4; all reported arithmetic is exact. The mathematical statement, the algebraic induction, and the boundary between proved and computational claims are presented explicitly above.

References

- [1] M. Omar, *New perspectives on the unimodality of domination polynomials*, arXiv:2601.14494v1 [math.CO] (2026). <https://arxiv.org/abs/2601.14494v1>
- [2] Z. Zeng, *The distinct-cycle-length probability is strictly decreasing after 30*, SSRN working paper 7380519 (2026). <https://doi.org/10.2139/ssrn.7380519>